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## **Rigid Robotic Transformations can approximate the kinematics of soft fingers with 'bones'**

*Hand Control and Dexterous Manipulation*

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### **Abstract:**

The ability to model and control soft bio-inspired robotic hands would enable a new class of manipulation applications because softness, by passively conforming, decreases the precision required for control (Brock and Valero-Cuevas, *Physics of Life Reviews*, 2016). Here we tested how well traditional rigid robotic transformations can approximate the kinematics of a semi-soft robotic finger (i.e., rigid 'phalanges' embedded in soft material). We made one tendon-driven, semi-soft finger with a length of 13 cm and three 4 cm phalanges (plus a 'metacarpal' for mounting) with 0.15 cm of silicone between serving as the 'joints'. Then, this semi-soft finger was compared against a ground-truth, rigid 3-link planar hinged finger.

Tendons were routed per the N+1 design (Valero-Cuevas, *Fundamentals of Neuromechanics*, 2016) where N is 3 degrees of freedom, in which tendons cross, and therefore affect multiple joints. Motors pulled on tendons with seven activation sets to drive the finger to different flexion-extension positions. The resulting finger endpoints were measured at each position using the DeepLabCut motion tracking software (Mathis et al., *Nature Neuroscience*, 2018). To test the validity of the linear rigid robotic transformations for our semi-soft finger, we calculated a linear regression relating endpoint locations to the seven tendon excursion sets.

The proportion of variance explained by the regression for the semi-soft finger was 69% ( $R^2=0.688$ ), compared to 100% for the rigid-finger. The average discrepancy between the predicted and observed finger endpoints for the 7 positions was  $2.1 (\pm 1.9)$  mm for the semi-soft finger and  $0 (\pm 0)$  mm for the rigid finger.

Our results indicate that while kinematic prediction of fingertip endpoints for the semi-soft finger did not follow the linear rigid robotic transformations as closely as that of the rigid finger, they were approximated well with about 69% of the variance explained. These results are encouraging as they show that there may be a way to combine the ability of semi-soft fingers to passively conform to objects grasped, with the numerous effective control methods developed for hinged rigid kinematics chains.

## References:

Brock and Valero-Cuevas. Transferring synergies from neuroscience to robotics: Comment on "hand synergies: Integration of robotics and neuroscience for understanding the control of biological and artificial hands" by m. santello et al. *Physics of Life Reviews*, 12(4):27–32, 2016. doi:10.1016/j.plrev.2016.05.011.

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